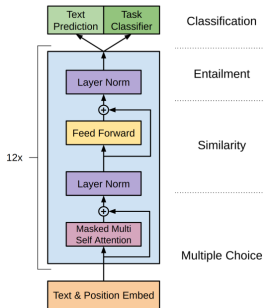


Deep Learning for NLP

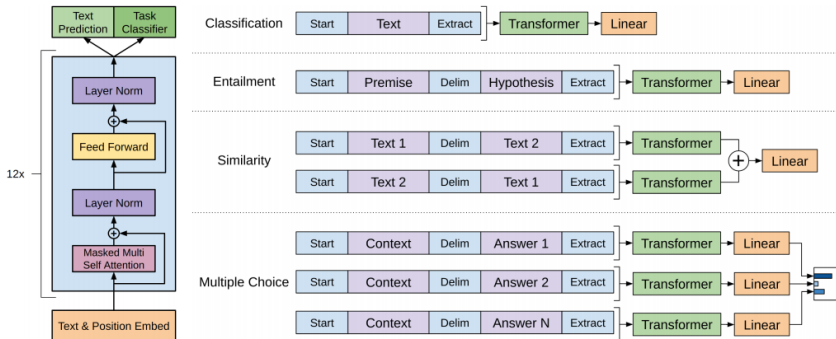
Generative Pre-Trained Transformers GPT-1 (2018)



Learning goals

- Use of the transformer decoder
- Input modifications (and how this is useful)

GPT-1



► Source: Radford et al., 2018

ARCHITECTURAL DETAILS

- Transformer *decoder* as backbone of the architecture
 - 12-layer-decoder with masked self-attention mechanism
 - Hidden dimension $H = 768$, $A = 12$ Attention heads
 - BPE vocabulary w/ 40k merges
 - Learned positional embeddings (as opposed to fixed, sinusoidal ones in the original Transformer)
- With $U = (w_{t-k}, \dots, w_{t-1})$

$$\vec{h}_0 = \vec{U}\vec{W}_e + \vec{W}_p$$

$$\vec{h}_l = \text{Trafo}(\vec{h}_{l-1}) \forall l \in [1, n]$$

$$P(w_t) = \text{softmax}(\vec{h}_n \vec{W}_e^\top)$$

PRE-TRAINING GPT

- Standard LM objective

$$L_1(\{w_1, \dots, w_n\}) = \sum_i \log(P(w_t | w_{t-k}, \dots, w_{t-1}; \Theta))$$

where $\{w_1, \dots, w_n\}$ is an *unlabeled* sequence of tokens

- *Resource*: BooksCorpus
 - > 7k unpublished books from various genres
 - contains long texts and thus allows learning long range dependencies

FINE-TUNING GPT

- Linear output layer with softmax activation on top
- Auxiliary language modeling objective during fine-tuning
 - Improves generalization
 - Accelerates convergence
- Task-specific input transformations
 - *Entailment*:
Concatenation of premise (p) & hypothesis (h): $[p; \$; h]$
 - *Similarity*: Use both orderings and concatenate resulting representations: $[s_1; \$; s_2]$ and $[s_2; \$; s_1]$
 - *Q&A and Commonsense Reasoning*:
Concatenate context (z), question (q) and each possible answer (a_k): $[z; q; \$, a_k]$
- Fine-tuning is rather quick, 3 epochs were sufficient

FINE-TUNING GPT

- Additional objective:

$$L_2(\{w_1, \dots, w_n\}) = \sum_{x,y} \log(P(y|w_1, \dots, w_n))$$

where

- $P(y|w_1, \dots, w_n) = \text{softmax}(h_l^m W_y)$ and
 - $\{w_1, \dots, w_n\}$ is a *labeled* sequence of tokens
- Combining both objectives:

$$L_3(\{w_1, \dots, w_n\}) = L_2(\{w_1, \dots, w_n\}) + \lambda \cdot L_1(\{w_1, \dots, w_n\})$$

SOTA RESULTS

Performance on different benchmarks:

Table 2: Experimental results on natural language inference tasks, comparing our model with current state-of-the-art methods. 5x indicates an ensemble of 5 models. All datasets use accuracy as the evaluation metric.

Method	MNLI-m	MNLI-mm	SNLI	SciTail	QNLI	RTE
ESIM + ELMo [44] (5x)	-	-	<u>89.3</u>	-	-	-
CAFE [58] (5x)	80.2	79.0	<u>89.3</u>	-	-	-
Stochastic Answer Network [35] (3x)	<u>80.6</u>	<u>80.1</u>	-	-	-	-
CAFE [58]	78.7	77.9	88.5	<u>83.3</u>		
GenSen [64]	71.4	71.3	-	-	<u>82.3</u>	59.2
Multi-task BiLSTM + Attn [64]	72.2	72.1	-	-	82.1	61.7
Finetuned Transformer LM (ours)	82.1	81.4	89.9	88.3	88.1	56.0

Table 3: Results on question answering and commonsense reasoning, comparing our model with current state-of-the-art methods.. 9x means an ensemble of 9 models.

Method	Story Cloze	RACE-m	RACE-h	RACE
val-LS-skip [55]	76.5	-	-	-
Hidden Coherence Model [7]	<u>77.6</u>	-	-	-
Dynamic Fusion Net [67] (9x)	-	55.6	49.4	51.2
BiAttention MRU [59] (9x)	-	<u>60.2</u>	<u>50.3</u>	<u>53.3</u>
Finetuned Transformer LM (ours)	86.5	62.9	57.4	59.0

► Source: Radford et al., 2018

GPT PARAMETER COUNT

- We know:

$$n_{layers} = 12; \quad d_{model} = 768; \quad V = 40000; \quad M = 512;$$

- Also:

$$N_{Decoder} = 12 \cdot d_{model}^2 \quad \text{and} \quad N_{Embedding} = \overbrace{V \times d_{model}}^{\text{token embedding}} + \underbrace{M \times d_{model}}_{\text{pos. embedding}}$$

$$\begin{aligned} \Rightarrow N_{total} &= n_{layers} \cdot N_{Decoder} + N_{Embedding} \\ &= 12 \cdot 12 \cdot 768^2 + 40000 \times 768 + 512 \times 768 \\ &= 116,047,872 \approx 117M \end{aligned}$$

Note that $N_{Decoder} = 12 \cdot d_{model}^2$ and not $16 \cdot d_{model}^2$ because the Decoder here doesn't do cross attention!